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Mitigating surface damage of probe pads in preparation for direct bonding of a substrate

Abstract

Mitigating surface damage of probe pads in preparation for direct bonding of a substrate is provided. Methods and layer structures prepare a semiconductor substrate for direct bonding processes by restoring a flat direct-bonding surface after disruption of probe pad surfaces during test probing. An example method fills a sequence of metals and oxides over the disrupted probe pad surfaces and builds out a dielectric surface and interconnects for hybrid bonding. The interconnects may be connected to the probe pads, and/or to other electrical contacts of the substrate. A layer structure is described for increasing the yield and reliability of the resulting direct bonding process. Another example process builds the probe pads on a next-to-last metallization layer and then applies a direct bonding dielectric layer and damascene process without increasing the count of mask layers. Another example process and related layer structure recesses the probe pads to a lower metallization layer and allows recess cavities over the probe pads.

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Background/Summary

RELATED APPLICATIONS (1) This patent application is a continuation of U.S. patent application Ser. No. 17/825,240, filed on May 26, 2022, which is a continuation of U.S. patent application Ser. No. 16/845,913, filed on Apr. 10, 2020, issued as U.S. Pat. No. 11,355,404, which is a non-provisional of U.S. Provisional Patent Application No. 62/837,004, filed on Apr. 22, 2019, the disclosures of which are hereby incorporated by reference herein, in their entireties.

BACKGROUND
(1) Test probing is routinely carried out on substrates, such as substrates of semiconductor materials, and on dies and reconstituted panels in microelectronics. Test probes make physical contact with probe pads on the given substrate. But the test probes can leave "probe marks" and surface disruption ("protrusions"), which can rise above the level of the dielectric passivation layer usually present around or above the probe pads. While this occurrence may not be a problem for some types of finished substrates, wafers, and dies, the probe pad protrusions can ruin the flatness of the overall top surface of the substrate for purposes of direct bonding processes that would join the substrate efficiently to other surfaces, for example in wafer-to-wafer bonding or die-to-wafer bonding, for example.
(2) Direct bonding processes include techniques that accomplish oxide-oxide direct-bonding between dielectrics, and also include techniques that accomplish hybrid bonding, which can bond metal interconnects together in an annealing step of the same operation that direct-bonds the dielectrics together.
(3) Conventional solutions for probe pad damage include adding sacrificial metallization layers or sacrificial probe pads to wafers, but these solutions are awkward and expensive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various technologies described herein.
(2) FIG. **1** is a diagram of a first example sequence of fabrication steps and resulting layer structures of an example wafer substrate, for mitigating damage to probe pads in preparation for direct bonding of the wafer substrate.
(3) FIG. **2** is a diagram of a continuation of the example sequence of fabrication steps and resulting layer structures of FIG. **1**, for mitigating damage to probe pads in preparation for direct bonding of the wafer substrate.
(4) FIG. **3** is a diagram showing various layer structures of a wafer substrate with direct bonding enabled on one or both surfaces of the wafer substrate by the example process shown in FIGS. **1-2**.
(5) FIG. **4** is a diagram of a second example sequence of fabrication steps and resulting layer structures on a wafer substrate, for mitigating damage to probe pads in preparation for direct bonding of the wafer substrate.
(6) FIG. **5** is a diagram of various embodiments of layer structures for a wafer substrate, in which probe pads are built in recessed cavities to prevent protrusions of disrupted probe pads from interfering with a direct bonding process at a direct bonding surface of the wafer substrate.
(7) FIG. **6** is a diagram of the wafer substrate of FIG. **5**, direct bonded to another wafer substrate, in which disrupted probe pads are isolated in recessed cavities that do not interfere with the direct bonding interface.

(8) FIG. **7** is a diagram of an example technique of using a liquid metal in recessed cavities built around probe pads to make electrical connections during probe testing while preventing disruption of the solid surface of the probe pads.
(9) FIG. **8** is a diagram of another example continuation of the sequence of fabrication steps and resulting layer structures of FIG. **1**, for mitigating damage to probe pads in preparation for direct bonding of the wafer substrate, in which interconnects or bonding pads on a flat surface to be direct bonded are conductively connected to damaged probe pads via a copper region in direct contact with the damaged probe pads.
(10) FIG. **9** is a flow diagram of an example method of preparing a wafer with probe pads for direct bonding after disruption of the probe pads, by filling-in and planarizing over the disrupted probe pads.
(11) FIG. **10** is a flow diagram of an example method of preparing a wafer with probe pads for direct bonding after disruption of the probe pads, by eliminating a top metallization layer of the wafer and substituting a direct hybrid bonding layer as top layer of the wafer.
(12) FIG. **11** is a flow diagram of an example method of preparing a wafer substrate with probe pads for direct bonding after disruption of the probe pads, by recessing the probe pads in recess cavities that isolate the probe pads from the direct bonding interface.
DETAILED DESCRIPTION
(13) Overview
(14) This disclosure describes methods and layer structures for mitigating surface damage of probe pads in preparation for direct bonding of a substrate, such as a reconstituted panel or the semiconductor substrate of a wafer or die.
(15) One example method prepares a semiconductor wafer for direct bonding processes by restoring a flat surface suitable for direct-bonding after disruption of probe pad surfaces during test probing. The example method fills a sequence of metals and oxides over the disrupted probe pad surfaces and builds out a dielectric surface and interconnects for hybrid bonding. An example layer structure associated with the method is described for increasing the yield and reliability of the resulting direct bonding process. Another example process builds the probe pads on a next-to-last metallization layer and then applies a direct bonding dielectric layer and a patterning or damascene process without increasing the count of metallization mask layers. Another example process and related layer structure recesses the probe pads to a lower metallization layer than the conventional top layer for probe pads, and forms recess cavities over the probe pads, which do not interfere with direct bonding at the topmost surface. In one case, liquid metals can be used in the recess cavities for test probing without disrupting probe pad surfaces.
(16) Example Processes and Layer Structures
(17) FIG. **1** shows a section of an example substrate, in this case a semiconductor wafer (or die) **100** to be joined to another wafer, die, or substrate in a direct bonding process. The wafer (or die) **100** is made of semiconductor material, such as silicon, and includes one or more layers of metallization **102** and probe pads **104** for testing the wafer (or die) **100**. The probe pads **104** may be aluminum metal (Al), copper (Cu), or other metal. The probe pads **104** may have protrusions **106** from contact with a test probe, which makes temporary contact during testing of the wafer (or die) **100**. The protrusions **106** may rise above the top level of a surrounding dielectric, oxide, or nitride layer **108**, preventing the top surface of the wafer (or die) **100** from attaining the flatness needed for direct bonding to another wafer or die.
(18) An example method deposits or overfills a layer or region of metal **110**, such as copper (Cu), over the probe pads **104** and over at least part of the protrusions **106**. The metal deposited may overfill **111** in a layer that covers the field regions **112** of the wafer (or die) **100**. In an implementation, an adhesion coating or a seed coating or a barrier coating of, for example, titanium (Ti), or Ta, or TaN, or TiN, or TiW (or a combination of these) may be deposited on at least the protrusions **106** and the probe pads **104** before the step of depositing or overfilling the metal **110**

over the probe pads **104**. The seed coating or barrier coating of Ti, Ta, TaN, TiN, or TiW, for example, may also cover larger areas of the field regions **112** prior to the step of depositing or overfilling the metal **110** over the probe pads **104** and potentially over these field regions **112** that may also have the barrier coating.

(19) As shown in FIG. **2** (continues FIG. **1**), the metal **110** deposited on the probe pads **104** and tops of the protrusions **106** are then planarized **114** by chemical mechanical polishing (CMP) or other polishing or flattening procedure to a flat surface **116** sufficient to meet a general planarization specification. The step of planarizing **114** both the metal **110** over the probe pads **104** and the protrusions **106** to the flat surface may include removing or polishing overfilled metal **111** on the field regions **112** of the wafer (or die) **100** to flatness until the metal **110** is removed from the field regions **112**.

(20) A layer of a dielectric material **118** is applied on the flat surface **116** provided by CMP. The layer of dielectric material **118** is a suitable material for direct bonding or hybrid bonding to another wafer, die, or substrate. In an implementation, the dielectric or oxide material **118** is a layer of "low temperature" oxide, applied by plasma enhanced chemical vapor deposition (PE-CVD), for example, such as a low temperature tetraethoxysilane (LT-TEOS), or another thermal oxide or other dielectric material suitable for direct bonding or hybrid bonding.

(21) The example process then creates a pattern in the layer of dielectric **118** using a damascene or other technique to make openings **120** over electrical contacts **122**, over through-silicon-vias (TSVs), or over other interconnects that are in contact with an underlying layer of metallization **102**.

(22) A metal **124** suitable for direct bonding is then deposited or plated in the openings **120** or in the pattern, to form interconnects. The deposited metal **124** may be prepared in various ways for the direct bonding process to occur at the topmost surface of the applied layers. In an implementation, a barrier layer of Ti, Ta, TaN, TiN, or TiW (or a combination of these) is deposited at least in the openings **120** or in both the openings **120** and the field regions **112** before depositing copper metal **124** or other metal in the openings **120** and on the field regions **112**, both of which may have a seed layer or barrier coating applied to their surfaces after the step of patterning the layer of dielectric **118**.

(23) The metal **124** and the layer of dielectric **118**, and the seed layer or barrier coating, when present, are then planarized with CMP or other technique to a flatness specification suitable for the direct bonding process or hybrid bonding process at the topmost surface **126**.

(24) In one example embodiment, the probe pads **104** are at least partially embedded in a layer of silicon nitride (Si.sub.3N.sub.4) or other dielectric. The metal **110** to be deposited, plated, or overfilled onto the probe pads **104** may be added up to a vertical height that reaches or fills-in to the top of a passivation layer, such as the silicon nitride or a silicon oxide layer **108**, around the probe pads **104**.

(25) FIG. **3** shows various example stack structures for substrates, such as wafers (or dies) in this example, made possible in various implementations by applying the example method described with respect to FIGS. **1-2**. The example stack structures enable direct bonding or hybrid bonding on one, or both, surfaces of the substrates, such as the wafer (or die) **100** and its built-up layers. The substrate may be part of a high bandwidth memory (HBM) wafer (or die) **100**.

(26) Example layer structure **302** for the wafer (or die) **100** provides a metal fill **110** and subsequent planarization **303** of disrupted probe pads **104** and the metal fill **110**. A layer of dielectric **118** on the top and bottom of the structure **302** allows dielectric-to-dielectric (oxide-oxide) direct bonding at the bottom surface, and hybrid bonding of both dielectric regions and metal regions on the top surface of the structure **302**. The oxide-oxide direct bonding (at the bottom surface) may be accomplished by oxide-to-oxide direct bonding, such as Zibond® brand direct bonding, for example (Xperi Corporation, San Jose, CA). The top surface provides a TSV reveal, with a surface possessing both metal regions and dielectric regions, enabled for hybrid bonding

such as can be accomplished by DBI® brand hybrid bonding (Xperi Corporation, San Jose, CA).

(27) Layer structure **304** has all the features of layer structure **302**, above, with the addition of interconnect metal **124** added in a hybrid bonding layer at the bottom of the structure **304**. In this example structure **304**, the interconnect metal **124** is electrically connected to the same circuit that the probe pad **104** is also connected to, thereby offering the possibility of test probing the circuit before connecting the same circuit to another wafer, die, or substrate.

(28) Layer structure **306** has all the features of the previous layer structure **304**, above, with the addition of a fuller fill of interconnect metal **124** on a bottom hybrid bonding layer. Thus, the same hybrid bonding layer that mitigates the protrusions **106** of the disrupted probe pads **104** is used as a full hybrid bonding layer with many interconnects. The interconnect metal **124** may be provided as a regular or non-regular array, pattern, or layout that includes operational pads and/or non-operational “dummy” pads.

(29) Layer structure **308** has all the features of the previous layer structure **306**, with the addition of the fuller fill of interconnect metal **124** on both top and bottom surfaces of the structure, which are both hybrid bonding surfaces, enabling full 3D wafer (or die) stacking on both sides of the wafer (or die) **100**, accomplished by hybrid bonding on both sides.

(30) The example layer structures **302-308** may have, on a first side of the wafer (or die) **100**, a probe pad **104** made of aluminum metal (Al) or copper metal (Cu) at least partially embedded in a layer of silicon nitride (Si.sub.3N.sub.4). A titanium (Ti) seed layer can be applied to at least protrusions **106** of the probe pad **104** caused by contact with a test probe. A copper region **110** is deposited on the titanium seed layer above the probe pad **104**. Then, a first silicon oxide layer **118** for direct bonding is applied on the surface of the wafer (or die) with an opening over the copper region **110**. Copper interconnects **124** for direct bonding are disposed through the silicon oxide layer **118** with at least some of the copper interconnects in contact with the copper region **110**.

(31) A tantalum (Ta) layer or other barrier layer material may also be applied between the copper region **110** and the silicon oxide layer **118**. The wafer **100** may be a high bandwidth memory (HBM) wafer **100**, including the vertical through-silicon vias (TSVs). The structures **302-308** may have other layers, such as another silicon oxide layer between the HBM wafer **100** and the layer of silicon nitride, and the silicon oxide layers **118** (or another dielectric) for direct bonding or hybrid bonding on one or more surfaces of the wafer **100**.

(32) FIG. **4** shows another layer structure and associated example method for preparing a substrate, such as a semiconductor wafer **400** for a direct bonding process after test probing the wafer **400** at the probe pads **104**. This example method eliminates a usual metallization layer, such as a conventional M4 layer, and instead fabricates a hybrid bonding layer in place of the eliminated conventional layer. The example method, besides solving the disruption of the probe pads **104**, results in no net increase in the number of mask layers for the wafer **400**, while adding the capability of hybrid bonding at a top surface.

(33) In greater detail, the example method includes creating the probe pads **104** in a metallization layer **402** that will underlie a top layer **404** for hybrid bonding to another wafer, die, or substrate. This example method builds the dielectric **406** of the top layer **404** directly over the probe pads **104**, whereas the example method associated with FIGS. **1-2** filled-in a metal **110** directly over the probe pads **104** instead of the dielectric **406**.

(34) After disruption of the probe pads **104** by a test probe resulting in protrusions **106**, the method deposits a layer of the dielectric **406**, such as silicon oxide, above the probe pads **104**. This layer of dielectric **406** is then planarized. Next, the layer of dielectric **406** is patterned to make openings over electrical contacts **408** in the underlying metallization layer **402**. The openings are filled with metal **410** to make interconnects **410** to be direct bonded during the direct hybrid bonding process.

(35) Atop surface of the interconnects **410** and the layer of dielectric **406** is planarized to a flatness specification suitable for hybrid bonding of the metal regions **410** and the nonmetal regions **406**, to another wafer, to a die, or to substrate.

(36) In an implementation, the thickness of the underlying metallization layer **402** with respect to the top layer **404** may be increased, since a conventional layer may be eliminated in this example method.
(37) FIG. **5** shows another example method for preparing a semiconductor wafer for a direct bonding process, and example layer structures associated with the method. In FIG. **5**, a conventional structure **500** shows the probe pads **104** on a topmost metallization layer of the wafer. The example method, by contrast, recesses the probe pads **104** further down from the top layer, within the underlying layers being fabricated on the wafer. The example method then leaves a recess cavity **510** above the probe pads **104**, thereby nullifying the effect of any protrusions **106** arising from the probe pads **104**, which may rise up with sufficient vertical height to interfere with a layer above the probe pads **104**, and may interfere with direct bonding to occur on the top layer, as would happen when the probe pads **104** are on top, as in the conventional structure **500**.
(38) The example method creates the probe pads **104** on a second-to-last metallization layer **502** or on a third-to-last metallization layer **504** of the wafer. After subsequent layers are built, the method creates a last layer **506** (topmost layer) including interconnects **508** compatible with a direct bonding process, the last layer **506** forming recess cavities **510** over the probe pads **104**.
(39) When the probe pads **104** are created on the third-to-last metallization layer **504**, the second-to-last metallization layer **502** also forms part of the recess cavities **510** over the probe pads **104**.
(40) In one variation, the example method includes increasing a thickness of the second-to-last metallization layer **502** or the third-to-last metallization layer **504** on which the probe pads **104** reside, with respect to a subsequent layer above.
(41) In FIG. **6**, when the wafer **506**, including at least the interconnects **508**, is direct bonded at interface **602** to another wafer **604**, die, or substrate, the recess cavities **510** are compatible with the direct bonding process, remaining open cavities **510** in one implementation.
(42) FIG. **7** shows an example structure depicting another example process, related to the method described with respect to FIGS. **5-6**. In FIG. **7**, a liquid metal **702**, such as gallium (Ga) may be placed in the recess cavities **510** to make an electrical contact between a test probe and a respective probe pad **104** without disrupting a metal surface of the probe pad **104**. This example method creates the same recess cavities **510** as described with respect to FIGS. **5-6**, but uses the recess cavities **510** to mitigate the problem of damaged probe pads **104**, albeit in a different manner. The method of FIGS. **5-6** allows the protrusions **106** of the probe pads **104** to exist, and just isolates the protrusions **106** in the recess cavity **510**, away from the direct bonding interface **602**. The method of FIG. **7** prevents damage to the probe pads **104** in the first place, by making test probe connections through the liquid metal **702**, ideally without contacting the solid surface of the probe pads **104**.
(43) FIG. **8** shows another process in continuation of the initial process steps shown in FIG. **1**. As in FIG. **1**, but referring to FIG. **8**, a metal **110** is deposited directly on the probe pads **104** and on top of the protrusions **106** or at least around the protrusions **106** caused by probe damage, and this metal **110** and the tops (if any) of the protrusions **106** undergo a planarization process **114** by chemical mechanical polishing (CMP) or another flattening or polishing procedure to become a flat surface **116** sufficient to meet a general planarization specification. The step of planarizing **114** applied to both the metal **110** over or around the probe pads **104** and the protrusions **106** to create the flat surface **116** may include removing or polishing overfilled metal on the field regions of the wafer **100** to flatness until the metal **110** is removed from those field regions.
(44) A layer of a dielectric material **118** is then applied on the flat surface **116** provided by CMP. The layer of dielectric material **118** is a suitable material for direct bonding or hybrid bonding to another wafer, die, or substrate. In an implementation, the dielectric or oxide material **118** is a layer of "low temperature" oxide, a low temperature tetraethoxysilane (LT-TEOS), or another dielectric material suitable for direct bonding or direct hybrid bonding as is known in the art.
(45) The example process then creates a pattern **120** & **800** in the layer of dielectric **118** using a

damascene or other technique to make openings **120** & **800** over electrical contacts **122**, over through-silicon-vias (TSVs), or over other interconnects that are in contact with an underlying layer of metallization **102**. In contrast to FIG. **2**, openings are also made (or only made) over the probe pads **104** and over the metal **110** that has been deposited and subjected to planarization **114**, over the probe pads **104**.
(46) A metal **124** & **802** suitable for direct bonding is then deposited or plated in the openings **120** & **800** or in the pattern, to form interconnects to be hybrid bonded at the direct bonding interface **804**. The deposited metal **124** & **802** may be prepared in various ways for the hybrid bonding process to occur at the topmost surface **804** of the applied layers. In an implementation, a barrier layer of Ti, Ta, TaN, TiN, or TiW (or a combination of these) can be deposited before depositing the metal **124** & **802**, such as copper or aluminum or another metal, in the openings **120** & **800** if intermixing of metal and semiconductor is an issue in a particular configuration, the barrier layer gets placed after patterning **120** & **800** the layer of dielectric **118**.
(47) The metal **124** & **802** and the layer of dielectric **118** is then planarized with CMP or other technique to a flatness specification suitable for the direct bonding process or hybrid bonding process at the topmost surface **804**.
(48) In one example embodiment, the probe pads **104** are at least partially embedded in a layer of silicon nitride (Si.sub.3N.sub.4) or other dielectric. The metal **110** to be deposited, plated, or overfilled onto the probe pads **104** may be added up to a vertical height that reaches or fills-in to the top of a passivation layer, such as a silicon nitride or a silicon oxide layer around the probe pads **104**.
(49) This example process of FIG. **8** provides a direct bonding surface **804** for hybrid bonding of the oxide layer **118** and the flat metal interconnects **124** & **802**, in which some of the interconnects **802** are conductively connected to the damaged probe pads **104** below via the deposited metal **110**. In an implementation, the interconnects **802** are only over the probe pads **104**, or only over the electrical contacts **122** of a metallization layer **102**, as in FIG. **2**.
(50) The hybrid bonding could also occur at flat surface **116** after planarization **114** and before placement of the dielectric layer **118** and interconnects **124** & **802**. In an embodiment, the bonding surface **108** may be plasma activated in preparation for bonding.
(51) If the wafer **100** is a substrate made of a dielectric or oxide material suitable for hybrid bonding, and not a semiconductor material, then the hybrid bonding may occur at surface **116** and one or more additional layers of metallization **118** & **124** & **802** may not be not needed.
(52) The implementations shown in FIG. **8** are compatible with the interconnect structures and layer configurations illustrated in FIG. **3**.
(53) Example Methods
(54) FIG. **9** shows an example method **900** of preparing a wafer with probe pads for direct bonding after disruption of the probe pads, by filling-in and planarizing over the disrupted probe pads. Operations of the example method **900** are shown in individual blocks.
(55) At block **902**, a wafer of semiconductor material is received, including at least one layer of metallization, and probe pads for testing the wafer. The probe pads may have protrusions and surface disturbances from contact with a test probe.
(56) At block **904**, a metal is deposited over the probe pads to cover at least part of the protrusions.
(57) At block **906**, the metal over the probe pads and protrusions is planarized into a flat surface.
(58) At block **908**, a layer of a dielectric is applied over the flat surface as a material for direct bonding.
(59) At block **910**, the layer of dielectric is patterned to make openings over electrical contacts in an underlying layer of metallization. The openings may be made by etching, a damascene process, or even by conventional via creation.
(60) At block **912**, a metal is deposited in the openings for making interconnects during direct bonding between the wafer and another wafer, or die, or substrate being direct bonded to.

(61) At block **914**, the metal in the openings, as well as the layer of dielectric is planarized to a flatness sufficient for direct bonding or direct hybrid bonding.
(62) FIG. **10** shows an example method of preparing a wafer with probe pads for direct bonding after disruption of the probe pads, by eliminating a top metallization layer of the wafer and substituting a hybrid bonding layer as top layer of the wafer. Operations of the example method **1000** are shown in individual blocks.
(63) At block **1002**, probe pads are created in an underlying metallization layer disposed beneath a top layer of a semiconductor wafer.
(64) At block **1004**, after disruption of the probe pads by a test probe, a layer of dielectric is deposited above the probe pads.
(65) At block **1006**, the applied layer of dielectric is planarized to flatness.
(66) At block **1008**, the layer of dielectric is patterned to make openings over electrical contacts in the underlying metallization layer.
(67) At block **1010**, the openings are plated or filled with a metal to make interconnects during direct bonding.
(68) At block **1012**, the top surface of the interconnects and the layer of dielectric are planarized to provide a flat surface with metal regions and nonmetal regions for direct bonding, such as hybrid bonding.
(69) FIG. **11** shows an example method of preparing a substrate, such as a wafer with probe pads for direct bonding after disruption of the probe pads, by recessing the probe pads in recess cavities that isolate the probe pads from the direct bonding interface. Operations of the example method **1100** are shown in individual blocks.
(70) At block **1102**, probe pads are created for a wafer substrate being fabricated for microelectronics by recessing the probe pads to a second-to-last or third-to-last metallization layer of the wafer substrate.
(71) At block **1104**, a last layer is created for the wafer substrate consisting of a dielectric and at least metal interconnects compatible with a direct bonding process, wherein the last layer has voids for forming recess cavities over the probe pads disposed on an underlying metallization layer.
(72) At block **1106**, when the probe pads are created on the third-to-last metallization layer, then the second-to-last metallization layer also forms part of the recess cavities over the probe pads.
(73) At block **1108**, the wafer is direct bonded to another wafer, a die, or a substrate, including direct bonding of the dielectric, and direct bonding of the interconnects. The recess cavities are compatible with the direct bonding process, and in one implementation remain open cavities after the direct bonding process.
(74) In the foregoing description and in the accompanying drawings, specific terminology and drawing symbols have been set forth to provide a thorough understanding of the disclosed embodiments. In some instances, the terminology and symbols may imply specific details that are not required to practice those embodiments. For example, any of the specific dimensions, quantities, material types, fabrication steps and the like can be different from those described above in alternative embodiments. The term “coupled” is used herein to express a direct connection as well as a connection through one or more intervening circuits or structures. The terms “example,” “embodiment,” and “implementation” are used to express an example, not a preference or requirement. Also, the terms “may” and “can” are used interchangeably to denote optional (permissible) subject matter. The absence of either term should not be construed as meaning that a given feature or technique is required.
(75) Various modifications and changes can be made to the embodiments presented herein without departing from the broader spirit and scope of the disclosure. For example, features or aspects of any of the embodiments can be applied in combination with any other of the embodiments or in place of counterpart features or aspects thereof. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

(76) While the present disclosure has been disclosed with respect to a limited number of embodiments, those skilled in the art, having the benefit of this disclosure, will appreciate numerous modifications and variations possible given the description. It is intended that the appended claims cover such modifications and variations as fall within the true spirit and scope of the disclosure.

Claims

1. A structure for direct bonding, including a buried probe pad in a substrate for microelectronics, the structure comprising: the probe pad formed in a probe pad metal layer of the substrate, the probe pad having an upper surface sized for test contact; an insulating material formed over the upper surface of the probe pad metal layer; a plurality of operational interconnects at least partially embedded in the insulating material; wherein the insulating material and the operational interconnects are part of a bonding surface prepared for direct hybrid bonding; and wherein the operational interconnects are electrically and physically connected to the probe pad metal layer through the insulating material without physically and electrically connecting through the upper surface of the probe pad.
2. The structure of claim 1, wherein the probe pad comprises at least one mark from probing the probe pad.
3. The structure of claim 2, wherein the at least one mark comprises at least one protrusion.
4. The structure of claim 2, further comprising a copper interconnect through the insulating material.
5. The structure of claim 2, wherein the probe pad metal layer comprises aluminum and the operational interconnects comprise copper.
6. The structure of claim 2, wherein the probe pad metal layer is a second-to-last metallization layer of the substrate.
7. The structure of claim 2, wherein the probe pad metal layer is a third-to-last metallization layer of the substrate.
8. The structure of claim 2, wherein the operational interconnects include a first operational interconnect on a first lateral side of the probe pad and a second operational interconnect on a second lateral side of the probe pad opposite the first lateral side.
9. The structure of claim 2, further comprising a second insulating material on an opposite side of the substrate from the insulating material; through-silicon vias (TSVs) in the substrate; and second interconnects through the second insulating material such that the second insulating material comprises a second direct hybrid bonding surface.
10. A bonded structure, including a buried probe pad in a first substrate for microelectronics, the structure comprising: the first substrate including the probe pad formed in a probe pad metal layer, the probe pad having an upper surface sized for test contact; a first insulating structure entirely covering the upper surface of the probe pad; a first operational interconnect for direct bonding, the first operational interconnect electrically connected through the first insulating structure to a first feature in the probe pad metal layer; and a second substrate including a second insulating structure and a second operational interconnect in the second insulating structure, wherein the second insulating structure is directly bonded to the first insulating structure of the first substrate, and the first operational interconnect is directly bonded to the second operational interconnect, such that an interface between the first and second substrates is a direct hybrid bonding interface.
11. The bonded structure of claim 10, wherein the probe pad comprises at least one mark from probing the probe pad.
12. The bonded structure of claim 11, wherein the at least one mark comprises at least one protrusion.
13. The bonded structure of claim 11, wherein the first feature in the probe pad metal layer is

positioned on a first lateral side of the probe pad, further comprising a third operational interconnect connected to a second feature in the probe pad metal layer through the first insulating structure, the second feature positioned on a second lateral side of the probe pad opposite the first lateral side, and the third operational interconnect being directly bonded to a fourth operational interconnect of the second substrate.

14. The bonded structure of claim 11, wherein the probe pad metal layer comprises aluminum.

15. The bonded structure of claim 14, wherein the first operational interconnect comprises copper and a barrier layer.

16. The bonded structure of claim 15, wherein the barrier layer comprises tantalum.

17. The bonded structure of claim 11, wherein the probe pad metal layer is a second-to-last metallization layer of the first substrate.

18. The bonded structure of claim 11, wherein the probe pad metal layer is a third-to-last metallization layer of the first substrate and at least a portion of the first operational interconnect is formed in a second-to-last metallization layer and a last metallization layer of the first substrate.

19. A semiconductor device for a direct hybrid bonding process, comprising: a probe pad in a metal layer of a semiconductor substrate, a contact surface of the probe pad comprising a mark characteristic of contact with a test probe, the contact surface being entirely covered by an insulating material; an insulating layer of the insulating material over the contact surface of the probe pad, the insulating layer including openings over electrical contacts in the metal layer adjacent to the probe pad; and a metal in the openings; wherein top surfaces of the metal and the insulating layer provide a direct hybrid bonding surface with metal and nonmetal regions for the direct hybrid bonding process.

20. The semiconductor device of claim 19, wherein the electrical contacts comprise a first electrical contact and a second electrical contact on opposite lateral sides of the probe pad.

21. The semiconductor device of claim 19, further comprising a second metal layer on the probe pad below the insulating layer.

22. The semiconductor device of claim 19, wherein the metal layer of the probe pad is a second-to-last metallization layer of the substrate, and the metal in the openings is a last metallization layer of the substrate.

23. The semiconductor device of claim 19, wherein the probe pad comprises a second metal composition different from a first metal composition in the openings.

24. The semiconductor device of claim 23, wherein the second metal composition comprises aluminum and the first metal composition in the openings comprises copper.

25. A method of preparing a semiconductor substrate for direct hybrid bonding, comprising: providing the semiconductor substrate, including a metal layer comprising a plurality of conductive features; testing circuitry of the semiconductor substrate including contacting a probe pad of the conductive features with a test probe; after testing, depositing at least one insulating layer above the conductive features; patterning the at least one insulating layer to form openings; providing metal in the openings to form operational bonding pads electrically connected to at least some of the conductive features without electrically connecting to the probe pad through an upper contact surface of the probe pad; and planarizing the metal and the at least one insulating layer to provide a surface having metal and nonmetal regions for direct hybrid bonding, the metal regions including the operational bonding pads.

26. The method of claim 25, further comprising depositing a metal material on the probe pad after testing and before depositing the at least one insulating layer.

27. The method of claim 25, wherein depositing the at least one insulating layer comprises depositing on a surface of the probe pad contacted by the test probe.

28. The method of claim 25, further comprising providing electrical connection through multiple insulating layers of the at least one insulating layer between the metal in the openings and the at least some of the conductive features.